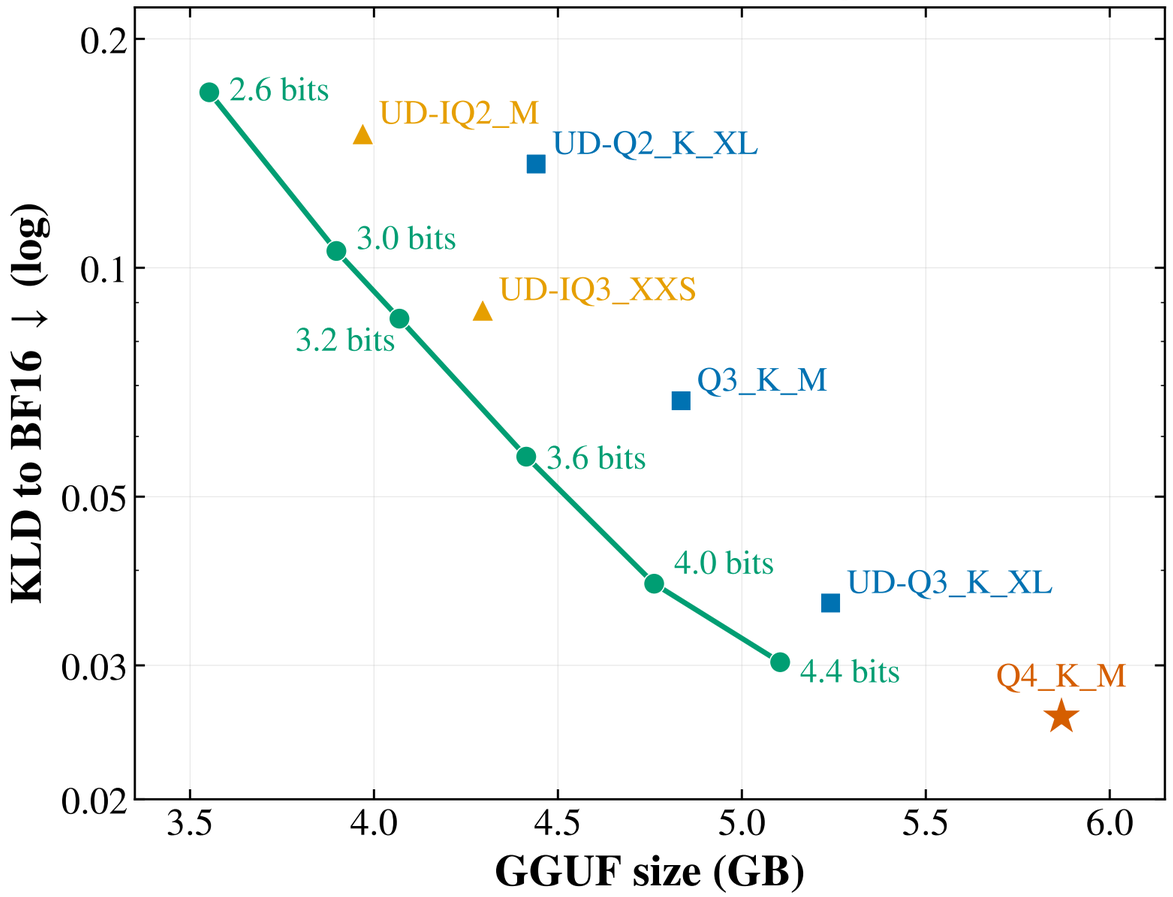
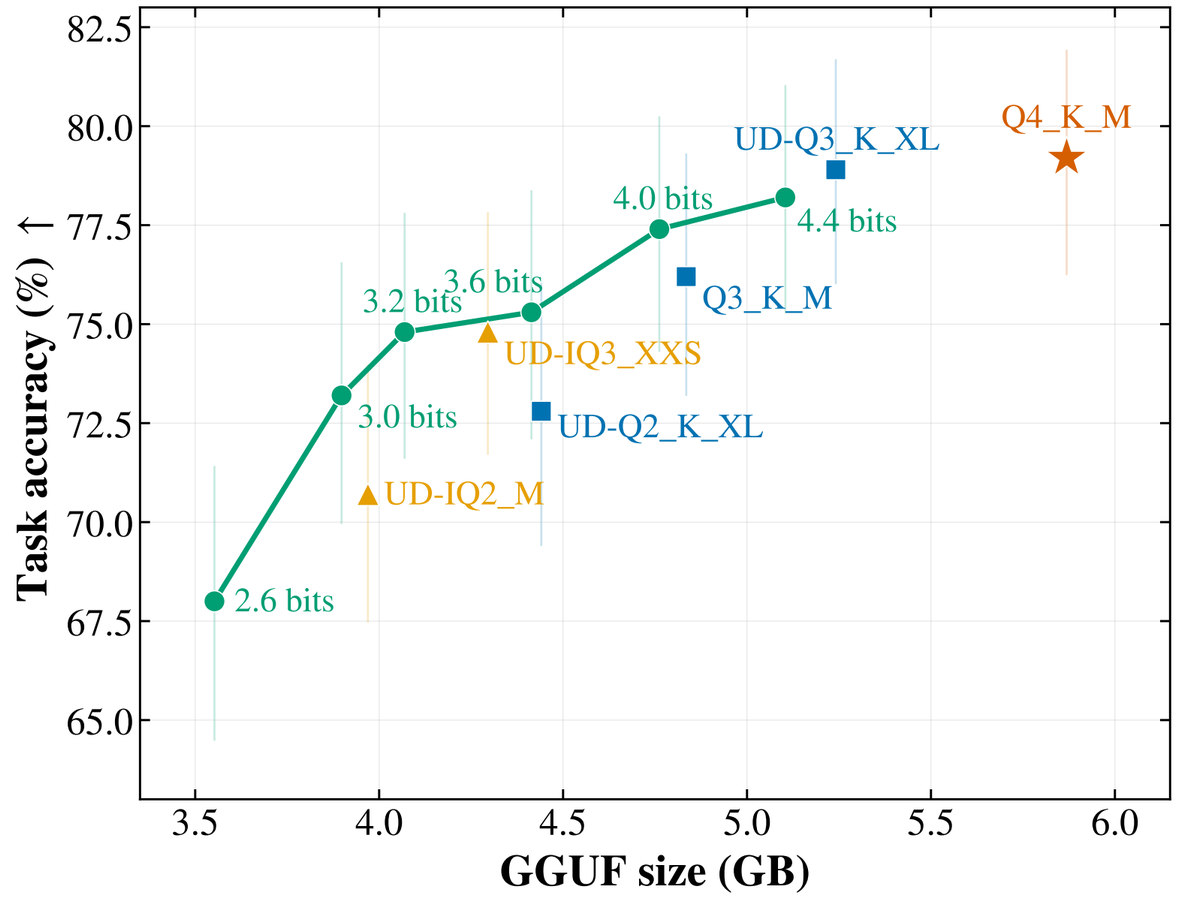
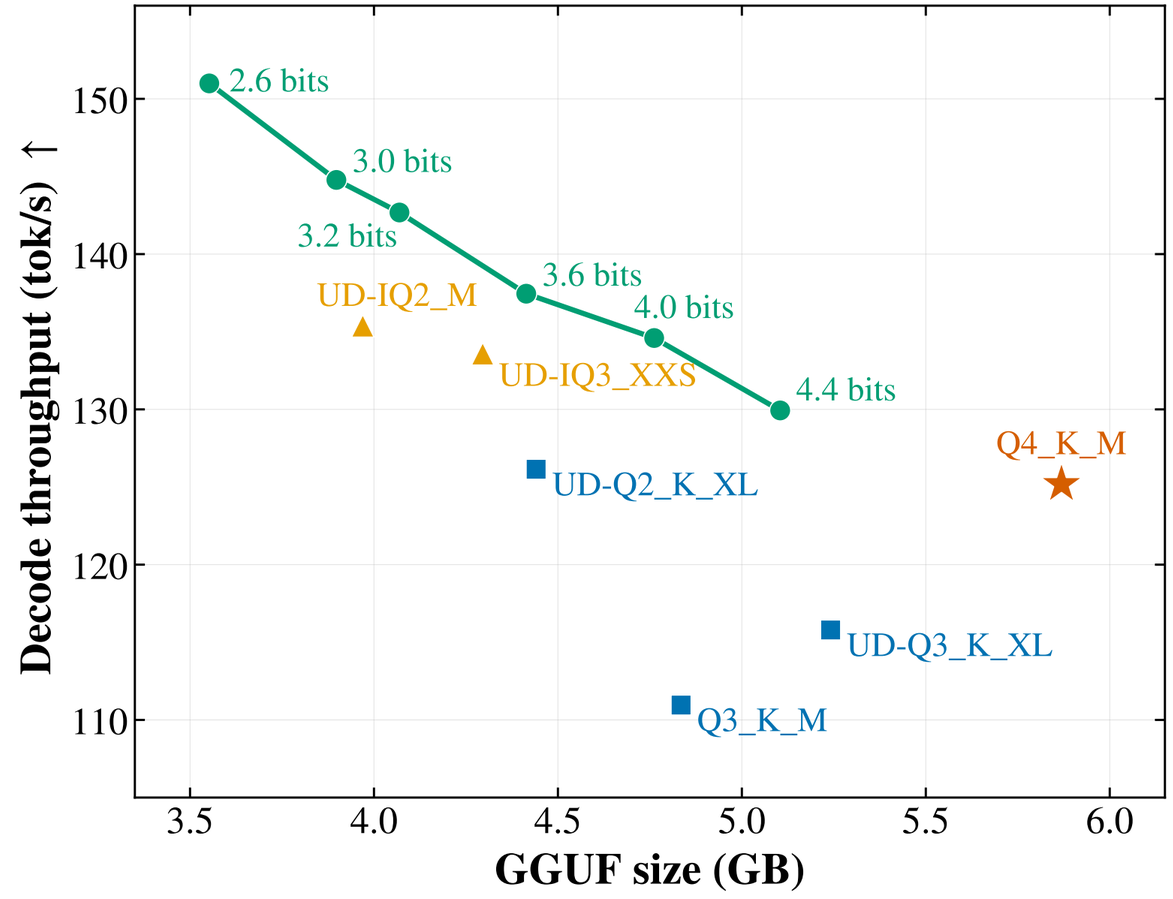
(a) Output fidelity



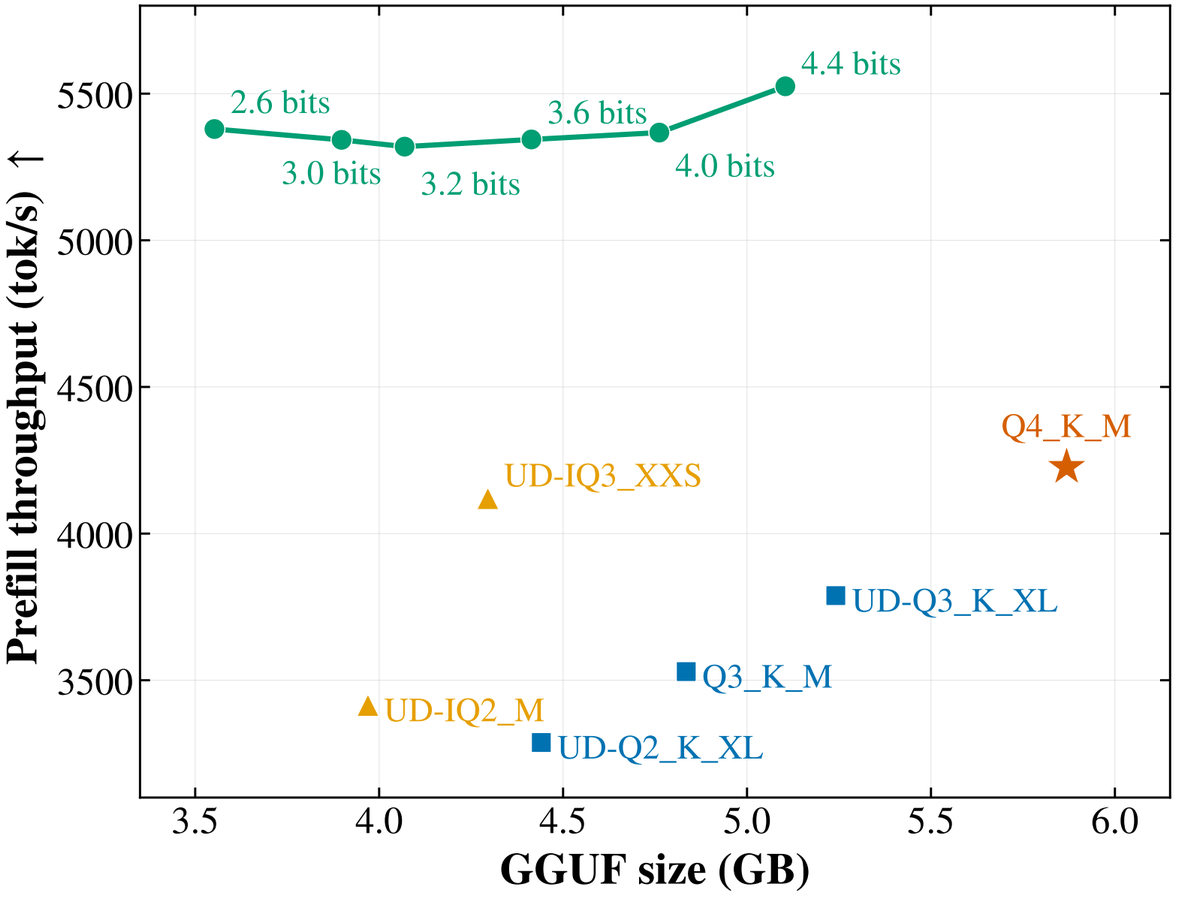
(b) Downstream tasks (95% CI)



(c) Decode speed (A100, CUDA)



(d) Prefill speed (A100, CUDA)



ShiftAddQwen-3.5-9B-*x*bits (ours)

UnslothAI/Qwen-3.5-9B-{Q3_K_M, UD-Q*_K_XL}

UnslothAI/Qwen-3.5-9B-Q4_K_M (reference)

UnslothAI/Qwen-3.5-9B-UD-IQ*